

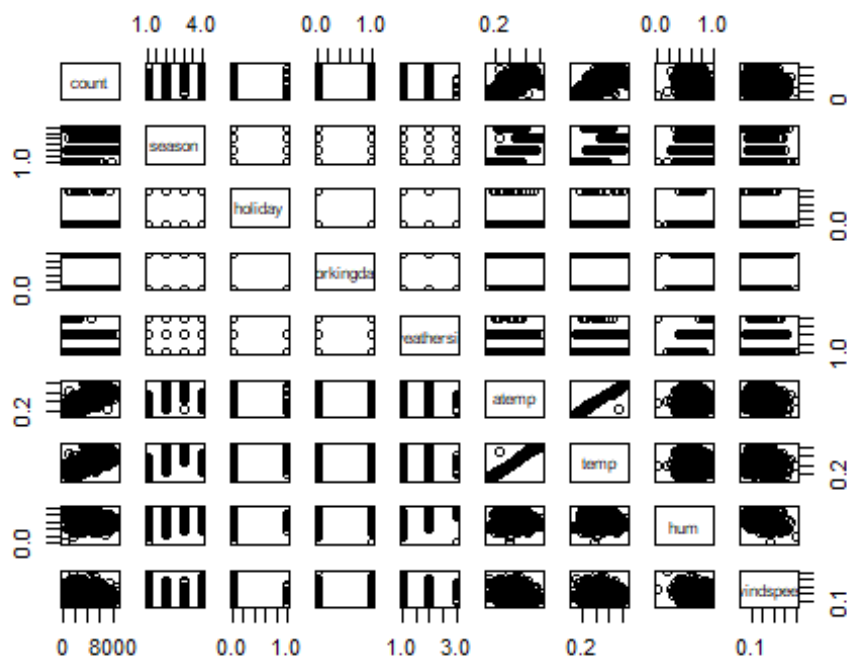
Final Analysis (54875)

Context: to predict daily bike rental count on the Enviromental and seasonal settings. And to give a brief analysis on the difference in the bike sharing analysis comparing holidays and with the working day.

Analysis

First I imported the dataset using the libraries `Alr4`, and `readr` and after importing the dataset I removed some of the variables which are not needed for the given analysis. They are `instant`, `dteday`, `yr`, `mnth`, `weekday` and even `causal` and `registered` because the sum of both is the count I think it's better to do the analysis removing this variable. And I visualized the data using the `pairs` function.

```
pairs(count~season+holiday+workingday+weathersit+atemp+temp+hum+windspeed,m0)
```



According to the marginal plots I see that season, holiday, workingday and weathersit are categorical variables. temp and atemp are linearly related to count and I can also see there is a strong correlation between atemp and temp. and humidity may be linearly related but that linearity depends on the leverage point also it's left asymmetric and likewise windspeed also has some influential points but it's right asymmetric. I have used the VIF function to find the collinear relation between the predictors.

So I got the strong correlation between the atemp and temp so I have removed the atemp and I have made a model using the remaining predictors.

As we have many categorical variables I have made them as a factor using the function as.factor

```
weather<-as.factor(m0$weathersit)
sea<-as.factor(m0$season)
holi<-as.factor(m0$holiday)
working<-as.factor(m0$workingday)
```

I have added this to my lm function. By removing the predictor atemp I have got a good model without the collinearity. Then I started analyzing the model using tests. I have started with residual plots

```
sea
holi
working
weather
temp      -9.6207      < 2.2e-16 ***
hum       -3.4517      0.0005896 ***
windspeed -0.3504      0.7261680
Tukey test -3.4641      0.0005320 ***
```

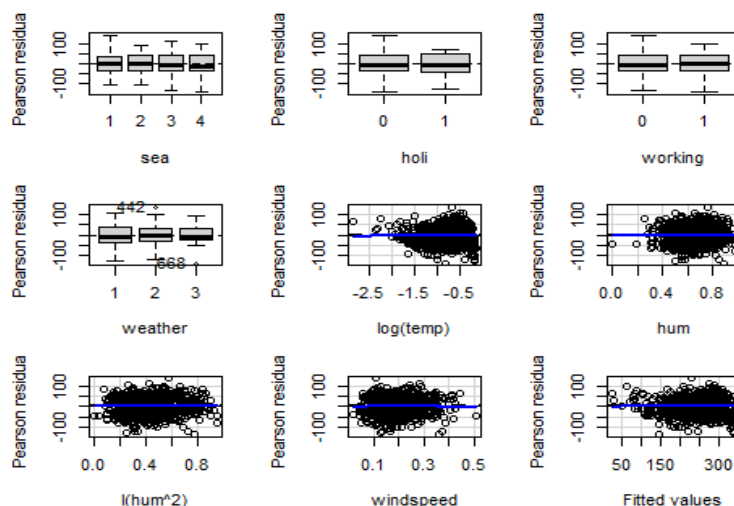
This is my previous results as we see that temp,hum and tukey test got significant p-values which means that we reject the null hypothesis which means that we have a curvature in our residual plot.so, I have tried different combinations of quadratic and interactions to the model. Most of the models doesn't work but I have found a model which is good.

```
m3<-lm(l(count^0.6592546 )~sea+holi+working+weather+log(temp)+hum+l(hum^2)+windspeed,m0)
```

the residual plot is this

```
Test stat Pr(>|Test stat|)
sea
holi
working
weather
log(temp)  -0.4773      0.6333
hum        -0.1390      0.8895
I(hum^2)    0.5755      0.5651
windspeed  -0.4250      0.6710
Tukey test  -0.1282      0.8980
```

As you can see that according to the test of the curvature I don't have evidence to reject the p-value which means that my model is good without the curvature.it doesn't mean our model is perfect.



Here in some of the plots we can see boxplots because of the categorical variables. Other than categorical variables we can see every plot does contain the quadratic pattern which is good for our model. Now let's test for the constant variance.

Non-constant Variance Score Test
 Variance formula: $\sim \text{fitted.values}$
 Chi-square = 26.19541, Df = 1, p = 3.0855e-07

When I did the constant variance test I got non constant variance because my p-value for the fitted value is 3.0855e-07 which is very low which states to reject the null hypothesis. So, our model contains nonconstant variance.

So here I tried doing the transformation for the y here is the result:
`powerTransform(count~sea+holi+working+weather+I(temp*windspeed)+hum+I(hum^2), m0)`

bcPower Transformation to Normality

	Est	Power	Rounded	Pwr	Wald	Lwr	Bnd	Wald	Upr	Bnd
y1	0.6593		0.66		0.5496		0.7689			

According to the summary I have got the estimated value for the response which is 0.6593 and the ci of upper limit and lower limit is (0.5496, 0.7698).

I have added the value to my main model. but now I have problem with the variable windspeed

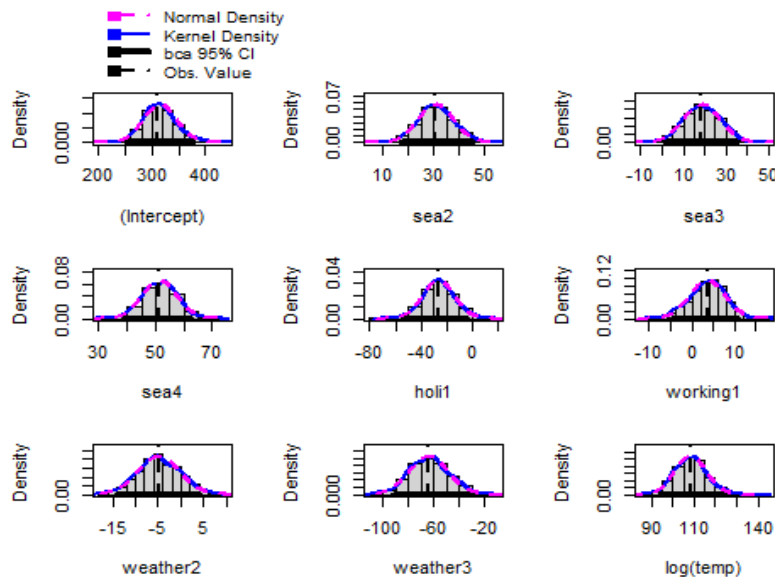
Non-constant Variance Score Test
 Variance formula: $\sim \text{windspeed}$
 Chi-square = 9.526654, Df = 1, p = 0.0020251

where the p-value is significant so we reject the null hypothesis. So there is no constant variance.

Finally, I tried to bootstrap

`B1<- Boot(m4,R=999)`

I have repeated the bootstrap sample 999 times. These are the results



The normality is good.

According to the summary,

	original	bootBias	bootSE	bootMed
## (Intercept)	310.9844	4.227625	31.3117	313.7801

## sea2	30.7909	0.141660	7.0067	30.7364
## sea3	18.5284	0.207983	8.7201	18.7967
## sea4	51.2997	0.136608	6.0957	51.6458
## holi1	-26.4332	0.472851	12.3521	-26.0439
## working1	3.5707	0.091220	4.1768	3.9214
## weather2	-4.8856	0.049766	4.6142	-4.8781
## weather3	-63.7226	-0.061891	15.3086	-63.9367
## log(temp)	108.3880	0.054757	7.4349	108.4569
## hum	232.3427	-14.462005	93.5853	225.0529
## I(hum^2)	-287.2946	11.016854	76.4813	-278.9424
## windspeed	-136.1542	0.797282	24.5796	-133.7388

this are my boot estimates the intercept is positive and its 310. And the remaining coefficient are follows: for season all coeff are positive which are (30.79,18.52,51.29) and for the holiday its negative and its -26.43 and follows boot bias values are the values with taking regressors into acc. Bootse is the standaed error boot vales and boomed is bootstrap median ones.

##	2.5 %	97.5 %
## (Intercept)	253.712500	375.302137
## sea2	17.476882	44.974956
## sea3	1.383929	34.739784
## sea4	38.854339	62.267679
## holi1	-53.507755	-3.142717
## working1	-5.147559	10.714670
## weather2	-14.054332	3.933631
## weather3	-92.707214	-32.928319
## log(temp)	93.868936	123.825191
## hum	36.267450	397.560537
## I(hum^2)	-432.146430	-141.007701
## windspeed	-188.996677	-91.173878

Conclusion:So according to the seasonality, our model has a good effect because CI doesn't include value 0 but we are not sure. The categories in the seasonality are significant for example sea2 coefficient is positive and it's 17.476882 so which means a 1 unit increase in the summer result in an increase in the count by 17.47 with respect to All the other variables. Here in the season, the reference model is when the season is springer. So we can compute the differences in the intercepts in the season using the reference one. Also using the weather I see that Light Snow, Light Rain + Thunderstorm + Scattered clouds, Light Rain + Scattered clouds does have an impact on the count because it doesn't include the value 0. The other Weather doesn't impact the count because it contains the 0 value. Our Bhat does include zero it's bad so we don't do CI as we know we do not have significance.

According to the difference between holidays and working days I see that holidays CI doesn't include zero which is better when compared to the model whether it includes the value 0 so it states at some point our sampling distribution passes through 0 which is not good for the analysis. So, its good in holidays comparing the working days.